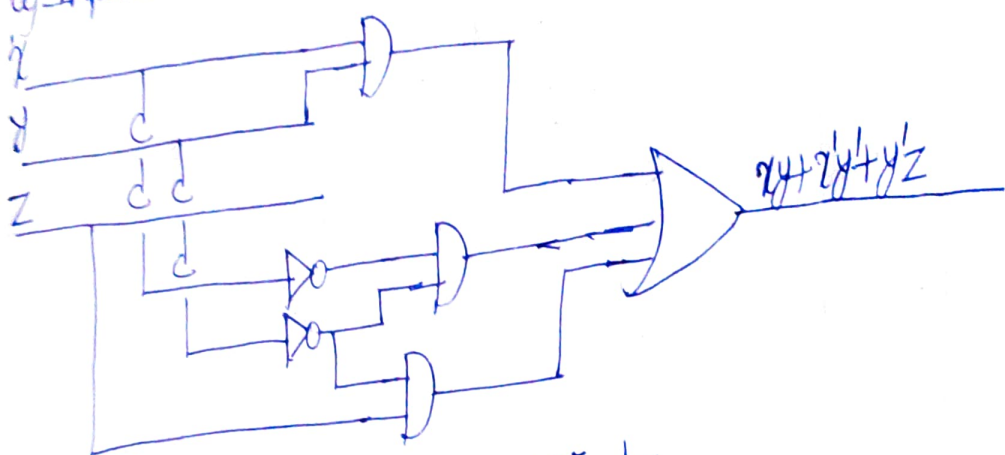


2-11: Given the Boolean function:

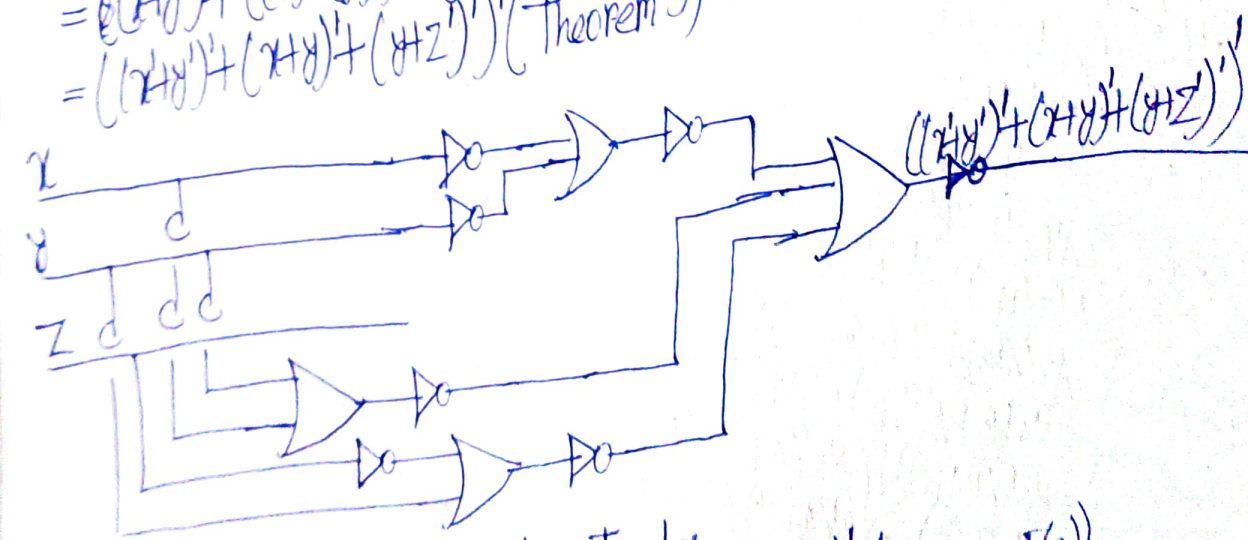
$$F = xy + x'y + y'z$$

a) Implement it with AND, OR and NOT gates



b) Implement it with only OR and NOT gates.

$$\begin{aligned} F &= xy + x'y + y'z = ((xy)')' + ((x'y)')' + ((y'z)')' \quad (\text{Theorem 3}) \\ &= ((x'y)')' + ((x' + y)')' + ((y' + z)')' \quad (\text{Theorem 5(b)}) \\ &= ((x'y)')' + (x + y)' + (y + z)' \quad (\text{Theorem 6}) \end{aligned}$$



c) Implement it with only AND and NOT gates.

$$F = xy + x'y + y'z = ((xy + x'y + y'z)')' = (x'y)' \cdot (x'y)' \cdot (y'z)' \quad (\text{Theorem 5(a)})$$